

Cybersecurity Incident Report:

Yummyrecipesforme.com

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

When a user attempts to visit [www.yummyrecipesforme.com, the browser sends a UDP packet to Port 53 to ask the DNS server for the website's IP address. In the captured network log, this lookup failed. Instead of returning the IP address, the network generated an ICMP error message originating from 203.0.113.2 and sent to the client machine at 192.51.100.15. The error explicitly noted that UDP Port 53 was unreachable.

Because Port 53 is the standard port reserved for DNS resolution, this error confirms that the request never reached an active DNS service. The issue is strictly isolated to the name resolution process rather than the web hosting service itself.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 1:24 PM

The IT team was alerted after multiple users reported that www.yummyrecipesforme.com would not load, displaying a "destination port unreachable" error in their browser.

Investigation Process:

To diagnose the outage, the IT department executed the following steps:

- Replicated the issue by navigating to the website.
- Initiated tcpdump to capture live packet data during a page refresh.
- Observed the browser initiate a standard DNS query over UDP to resolve the domain name.
- Analyzed the resulting tcpdump log, which captured ICMP error responses triggered by the outgoing UDP packets to Port 53.

Potential Causes:

The log data confirms that client traffic cannot complete DNS requests on Port 53. The

root cause falls into one of three likely categories:

Firewall Restrictions: A network firewall or security rule is blocking inbound UDP traffic on Port 53.

Service Disruption: The DNS service daemon crashed or is unconfigured on the target host.

Denial of Service (DoS): An influx of malicious traffic overloaded the DNS infrastructure, rendering it unresponsive to legitimate queries.